

Engineering Statistics

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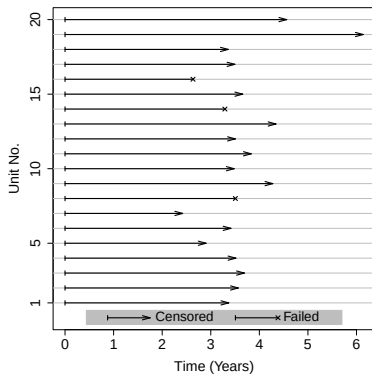
Chapter 14: Reliability and Survival Analysis

From linear errors to time-to-event data

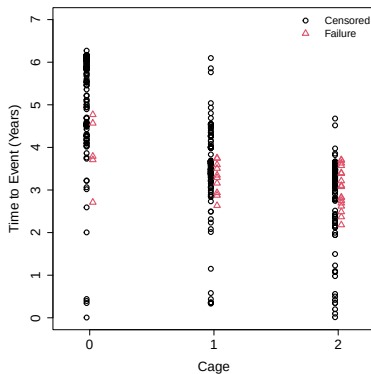
- ▶ Classical setting: response Y with model $Y_i = \mathbf{x}_i^\top \beta + \sigma \epsilon_i$ with $\mathbb{E}\epsilon_i = 0$, $\text{Var}(\epsilon_i) = 1$.
- ▶ In reliability/survival studies, the response is a *time to event* $T > 0$:
 - ▶ positive support, often strongly right-skewed;
 - ▶ frequently *right-censored*.
- ▶ Normal-error linear models are not suitable for T directly.

Example: Titan GPU OTB data

- ▶ Response: time in service until OTB failure (event) or censoring.
- ▶ Covariate: *Cage* (server cabinet position).



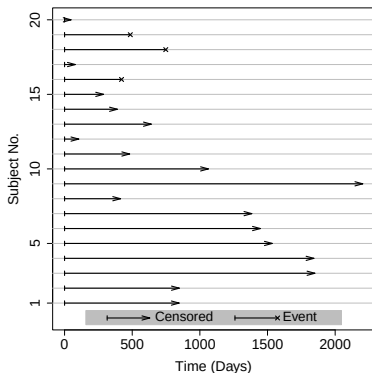
(a) Event & censoring indicators



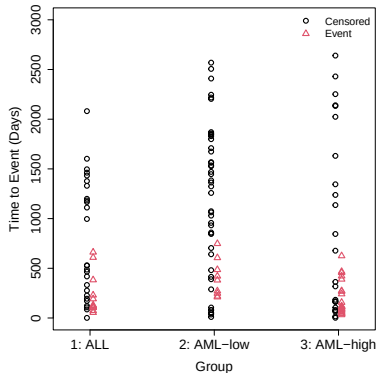
(b) Time vs. Cage

Example: Bone Marrow Transplant (BMT)

- ▶ Response: survival time; many right-censored observations.
- ▶ Covariate: Disease Group (ALL, AML-low, AML-high).



(a) Event plot



(b) Time vs. Disease Group

Right censoring

Right censoring: Let T be the event time and t_0 the last time the unit is observed. If $T > t_0$, the observation is *right-censored*.

- ▶ Data are triples $(t_i, \delta_i, \mathbf{x}_i)$ with

$$\delta_i = \begin{cases} 1, & \text{event observed at } t_i, \\ 0, & \text{right-censored at } t_i. \end{cases}$$

- ▶ Censoring is *not* missing: it provides a lower bound for T .

Work with $\log T$ (location–scale family)

Let $Y = \log T$. Model Y with a location–scale family:

$$F_Y(y) = \Phi\left(\frac{y - \mu}{\sigma}\right), \quad f_Y(y) = \frac{1}{\sigma} \phi\left(\frac{y - \mu}{\sigma}\right).$$

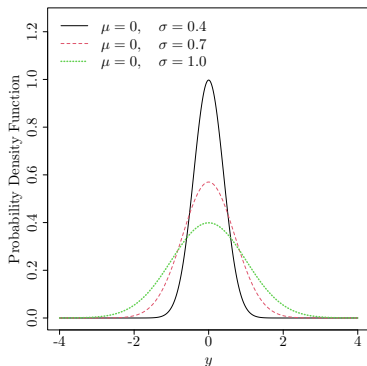
Log-location–scale: If Y follows a location–scale distribution, then $T = \exp(Y)$ follows a *log-location–scale* distribution.

- ▶ Common Y choices and induced T :
 - ▶ Normal $\Rightarrow$ Lognormal,
 - ▶ Smallest Extreme Value (SEV) $\Rightarrow$ Weibull,
 - ▶ Largest Extreme Value (LEV) $\Rightarrow$ Frechet,
 - ▶ Logistic $\Rightarrow$ Loglogistic.

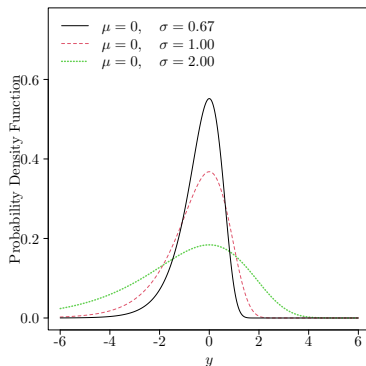
Standard families (for $Y = \log T$)

T	$Y = \log T$	pdf	cdf
Lognormal	Normal	$\phi_N(z) = (2\pi)^{-1/2} e^{-z^2/2}$	$\Phi_N(z) = \int_{-\infty}^z \phi_N(u) du$
Weibull	SEV	$\phi_{SEV}(z) = \exp(z - \exp z)$	$\Phi_{SEV}(z) = 1 - \exp(-\exp z)$
Frechet	LEV	$\phi_{LEV}(z) = \exp(-z - \exp(-z))$	$\Phi_{LEV}(z) = \exp(-\exp(-z))$
Loglogistic	Logistic	$\phi_{LGS}(z) = \frac{e^z}{(1 + e^z)^2}$	$\Phi_{LGS}(z) = \frac{e^z}{1 + e^z}$

Pdf illustrations (Normal vs. SEV)



(a) Normal



(b) SEV

AFT model and induced distribution for T

AFT model

$$Y_i = \log T_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \sigma \epsilon_i,$$

with ϵ_i from a standard location–scale distribution.

Let $\mu(\mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta}$. Then

$$F(t; \mathbf{x}) = \Pr(T \leq t) = \Phi\left(\frac{\log t - \mu(\mathbf{x})}{\sigma}\right), \quad (1)$$

$$f(t; \mathbf{x}) = \frac{1}{t\sigma} \phi\left(\frac{\log t - \mu(\mathbf{x})}{\sigma}\right). \quad (2)$$

Acceleration factor and cdf rescaling

Fix a baseline covariate $\mathbf{x}_0$ and define

$$AF(\mathbf{x}) = \exp\{\mu(\mathbf{x}_0) - \mu(\mathbf{x})\} = \exp\{(\mathbf{x}_0 - \mathbf{x})^\top \boldsymbol{\beta}\}. \quad (3)$$

Then, using (1),

$$F(t; \mathbf{x}) = \Phi\left(\frac{\log t - \mu(\mathbf{x})}{\sigma}\right) = \Phi\left(\frac{\log(AF(\mathbf{x}) t) - \mu(\mathbf{x}_0)}{\sigma}\right) = F(AF(\mathbf{x}) t; \mathbf{x}_0). \quad (4)$$

If $AF(\mathbf{x}) > 1 \Leftrightarrow \mathbf{x}_0^\top \boldsymbol{\beta} > \mathbf{x}^\top \boldsymbol{\beta}$ the time scale is accelerated relative to baseline; if < 1 it is decelerated.

Likelihood with right censoring: contributions

For each $(t_i, \delta_i, \mathbf{x}_i)$ let

$$z_i = \frac{\log t_i - \mathbf{x}_i^\top \boldsymbol{\beta}}{\sigma}.$$

- ▶ Event ($\delta_i = 1$): contribution $f(t_i; \mathbf{x}_i)$ from (2).
- ▶ Right-censored ($\delta_i = 0$): contribution $S(t_i; \mathbf{x}_i) = 1 - F(t_i; \mathbf{x}_i)$ via (1).

Full likelihood and parameters

The full likelihood is

$$\mathcal{L}(\boldsymbol{\theta}) = \prod_{i=1}^n \left(\frac{1}{\sigma t_i} \phi(z_i) \right)^{\delta_i} \left(1 - \Phi(z_i) \right)^{1-\delta_i}, \quad (5)$$

with $\boldsymbol{\theta} = (\boldsymbol{\beta}^\top, \sigma)^\top$.

- ▶ Use baseline parameterization for categorical covariates.
- ▶ Numerical maximization of $\log \mathcal{L}$.

Estimation: Newton-type optimization and information

- ▶ Maximize $\log \mathcal{L}(\theta)$ in (5) using a Newton-type method.
- ▶ Observed (local) information at $\hat{\theta}$:

$$\mathbf{I}(\hat{\theta}) = -\frac{\partial^2}{\partial \theta \partial \theta^\top} \log \mathcal{L}(\theta) \Big|_{\theta=\hat{\theta}}.$$

- ▶ Estimated covariance: $\mathbf{I}(\hat{\theta})^{-1}$.

Large-sample inference for AFT

Under mild regularity,

$$\hat{\boldsymbol{\theta}} \rightsquigarrow \mathcal{N}(\boldsymbol{\theta}, \mathbf{I}(\hat{\boldsymbol{\theta}})^{-1}).$$

- CIs and tests for β_j and σ follow.

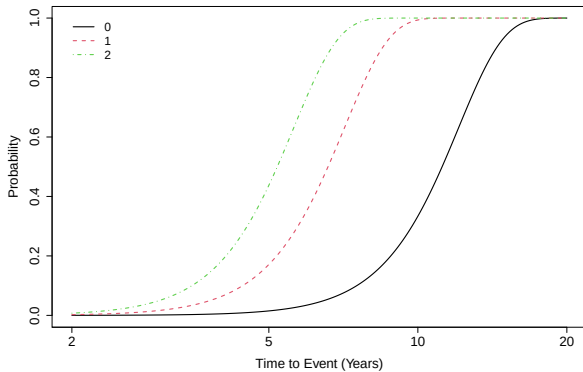
AFT Example: GPU data (SEV errors)

Baseline: Cage 0. 95% CIs via large-sample theory.

Parameter	Est	SE	CI Lower	CI Upper
β_0	2.493	0.145	2.209	2.776
Cage1	-0.531	0.118	-0.763	-0.299
Cage2	-0.767	0.113	-0.989	-0.544
σ	0.211	0.032	0.157	0.282

Acceleration factors from (3): $AF(C1) = e^{0.531} = 1.70$,
 $AF(C2) = e^{0.767} = 2.15$.

Estimated cdf curves (GPU)



Hazard, CHF, and survival

Hazard function:

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{\Pr(t < T \leq t + \Delta t \mid T > t)}{\Delta t} = \frac{f(t)}{S(t)}.$$

Cumulative hazard function (CHF): $H(t) = \int_0^t h(s) ds$.

We have $S(t) = \exp\{-H(t)\}$.

Proof sketch: $h(t) = -\frac{d}{dt} \log S(t)$; integrate from 0 to t .

Cox PH model and hazard ratio

Model (no intercept)

$$h(t; \mathbf{x}) = h_0(t) \exp(\mathbf{x}^\top \boldsymbol{\beta}),$$

with baseline at $\mathbf{x} = \mathbf{0}$.

For two covariate vectors $\mathbf{x}_1, \mathbf{x}_2$,

$$\text{HR}(\mathbf{x}_1, \mathbf{x}_2) = \frac{h(t; \mathbf{x}_1)}{h(t; \mathbf{x}_2)} = \exp\{(\mathbf{x}_1 - \mathbf{x}_2)^\top \boldsymbol{\beta}\}, \quad (6)$$

which is time-invariant (“proportional hazards”).

Risk sets and event ordering

Let $t_1 < \dots < t_f$ be the f distinct event times (assume no ties).

- ▶ $\mathcal{R}(t_j) = \{i : t_i \geq t_j\}$: indices of individuals at risk just before t_j (not yet failed nor censored before t_j).
- ▶ Those who had event/censoring exactly at t_j are elements of $\mathcal{R}(t_j)$.
- ▶ Only ordering of event times matters.

Partial likelihood (derivation)

The probability individual i_j is the one who fails at t_j (conditioned on $\mathcal{R}(t_j)$) is

$$\begin{aligned} L_j(\beta) &= \frac{\Pr(\text{individual } i_j \text{ failed at } t_j \mid n_j \text{ survived up to } t_j)}{\Pr(\text{one of the } n_j \text{ failed at } t_j \mid n_j \text{ survived up to } t_j)} \\ &\approx \frac{h(t_j; \mathbf{x}_{i_j})}{\sum_{i \in \mathcal{R}(t_j)} h(t_j; \mathbf{x}_i)} = \frac{h_0(t_j) \exp(\mathbf{x}_{i_j}^\top \beta)}{\sum_{i \in \mathcal{R}(t_j)} h_0(t_j) \exp(\mathbf{x}_i^\top \beta)} = \frac{\exp(\mathbf{x}_{i_j}^\top \beta)}{\sum_{i \in \mathcal{R}(t_j)} \exp(\mathbf{x}_i^\top \beta)}. \end{aligned} \quad (7)$$

The *partial likelihood* is

$$\mathcal{L}(\beta) = \prod_{j=1}^f L_j(\beta). \quad (8)$$

Baseline hazard $h_0(t)$ cancels in $L_j(\beta)$, so it need not be specified.

Estimation and large-sample inference (Cox)

- ▶ Maximize $\log \mathcal{L}(\beta)$ in (8) numerically (Newton-type).
- ▶ Asymptotically, $\hat{\beta}$ is normal with covariance from the observed information of the partial likelihood.

Baseline CHF and survival (Breslow)

Given $\hat{\beta}$, the baseline CHF is

$$\hat{H}_0(t) = \sum_{j: t_j \leq t} \frac{1}{\sum_{\ell \in \mathcal{R}(t_j)} \exp(\mathbf{x}_\ell^\top \hat{\beta})}. \quad (9)$$

For covariate $\mathbf{x}$:

$$\hat{H}(t; \mathbf{x}) = \exp(\mathbf{x}^\top \hat{\beta}) \hat{H}_0(t), \quad \hat{S}(t; \mathbf{x}) = \exp\{-\hat{H}(t; \mathbf{x})\}.$$

$\hat{H}_0(t)$ is a step function with jumps at event times.

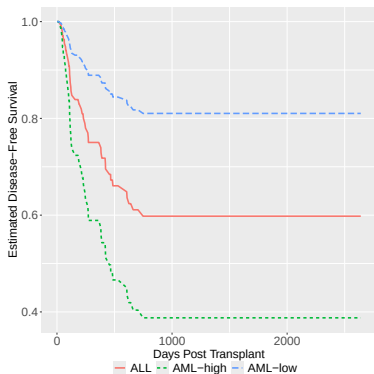
Cox PH Example: BMT (disease groups)

Baseline = ALL; indicators for AML-low and AML-high.

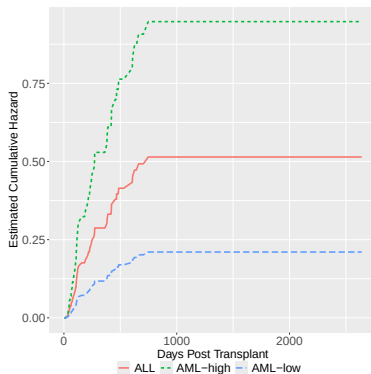
Parameter	Estimate	$\exp(\beta)$	Std Err	z	p
AML-low	-0.894	0.409	0.442	-2.022	0.043
AML-high	0.611	1.842	0.362	1.686	0.092

Interpretation: relative to ALL, AML-high has higher hazard, AML-low lower hazard.

Estimated survival and CHF (BMT)



(a) Survival



(b) CHF

Key points

- ▶ AFT: $Y = \log(T) = \mathbf{x}^\top \beta + \sigma \epsilon$; Y is from location scale families; cdf $F(t; \mathbf{x})$ in (1); acceleration factor (3) rescales time via (4).
- ▶ Likelihood with censoring: product $f(t)^\delta S(t)^{1-\delta}$ (see (5)); MLE via Newton-type optimization; asymptotic normality for inference.
- ▶ Cox PH: $h(t; \mathbf{x}) = h_0(t) \exp(\mathbf{x}^\top \beta)$; hazard ratios are time-invariant; estimate β by partial likelihood (8); baseline CHF via Breslow (9).
- ▶ Examples: GPU (AFT) and BMT (Cox) illustrate modeling and interpretation.

Application: Reliability of AI Systems

- ▶ **AI systems can fail.** Incorrect decisions, unsafe actions, and failures under changing conditions.
- ▶ **Define and collect failure data.** Record failure events and time to failure.
- ▶ **Apply reliability and survival analysis.** Model failure probabilities using survival and hazard functions.
- ▶ **Address AI-specific challenges.** Failures depend on inputs, environments, and distribution shifts.
- ▶ **Improve AI system reliability.** Use failure analysis to guide testing and monitoring.

